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CONTRIBUTION**Source:** India**Title:** Proposals for extension of ITU-T P.1503 (ex P.DFSInter): Extended methodology for cross-country and inter-operator digital financial services testing**Contact:** Venkata Rama Raju Chelle
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E-mail: cvramaraju.v@gov.in**Abstract:** This document proposes to extend Recommendation ITU-T P.1503 (ex P.DFSInter) to incorporate the Unified Payment Interface (UPI) developed in India, extended with global acceptance on inclusion basis to assess QoS and QoE of each payment service provider.**Draft New Recommendation ITU-T P.DFS_Inter****Extended methodology for cross-country and inter-operator Digital Financial Services testing to assess QoS and QoE of each payment service provider****Summary**

This work is based on ITU-T Rec. G.1033, where a conceptual framework for Quality of service and quality of experience aspects of digital financial services is standardised, and on ITU-T Rec. P.1502 which standardizes a methodology for QoE testing of digital financial services for the basic person-to-person (P2P) money transfers between two devices using the same network and DFS operator. This is extended beyond user and one network. that powers multiple bank accounts into a single mobile application (of any participating bank), merging several banking features, seamless fund routing & merchant payments into one hood. It also caters to the “Peer to Peer” collect request which can be scheduled and paid as per requirement and convenience.

- Immediate money transfer through mobile device round the clock 24*7 and 365 days.
- Single mobile application for accessing different bank accounts.

- Single Click with 2 Factor Authentication – Aligned with the Regulatory guidelines, yet provides for a very strong feature of seamless single click payment.
- Virtual address of the customer for Pull & Push provides for incremental security with the customer not required to enter the details such as Card no, Account number; IFSC etc.
- QR Code or Mobile number (linked with virtual bank address)
- Best answer to Cash on Delivery hassle, running to an ATM or rendering exact amount.
- Merchant Payment with Single Application or In-App Payments.
- Utility Bill Payments, Over the Counter Payments, QR Code (Scan and Pay) based payments.
- Donations, Collections, Disbursements Scalable.
- Raising Complaint from Mobile App directly.

Participants elements in the UPI

- Payer PSP
- Payee PSP
- Remitter Bank
- Beneficiary Bank
- NPCI (gateway provider)
- Bank Account holders
- Merchants/Business houses/Enterprises

The new Recommendation will have three main elements. Firstly, the methodological framework and use case definitions for a generalized P2P or P2M money transfer use case will be given. In this framework, the DFS operator i.e., PSP used to send money, and the operator receiving this very money (i.e., the Remitter and Merchant/Receiver side of a money transfer) are parameters of the use case, which integrates all variations (same PSP/inter-PSP; same country/cross-country) into the same methodological context.

The second element of the Recommendation will be a comprehensive framework for data elements and related processing, including tools and procedures providing operational robustness and a high level of data quality. The data objects defined in the new Recommendation will support test planning and management as well as provide the input data foundation for efficient processing of data. Also, guidance will be given on how data processing can be done in a consistent and efficient way.

Last but not least, the Recommendation will introduce a new tool, and the corresponding methodology, designed to assist field test teams in data collection. This tool (“multi-stopwatch”) has already been conceptually suggested in ITU-T Rec. P.1502. It is an electronic time-taking tool similar to a stopwatch but supporting testers to record the events within a DFS test case and upload data entered by testers directly to a central location. This tool eliminates the needs and weaknesses of manual entering of time readings and the process of transferring them to post-processing through multiple transformation stages, e.g. from handwritten notes to entries in a spreadsheet.

Keywords

UPI, VPA, PSP, BHIM, Remitter, Merchant bank, Recipient bank, NPCI, UPI

Introduction

The Unified Payment Interface (UPI) provides a single mobile application interface that allows seamless interoperability between different payment systems.

UPI works on the concept of a virtual payment address. Bank accounts, cards and wallets can be mapped to a unique virtual payment address. Payments can be made using an account number, mobile number and Aadhaar number (virtual payment address). UPI leverages the existing infrastructure for authentication.

1. Scope

This Recommendation describes the QoE assessment methodology for the "Person-to-Person" (P2P) and **Peer-to-Merchant** for money transfer or payment to merchant use case, i.e. from party A to party B. It covers the full range of possible combinations of operator of digital financial services (DFS) of PSP and country where the parties are located, i.e. for the case where A and B are using the same operator or different ones, as well as intra-country and cross-country transfers. Therefore, this Recommendation is an extension of Recommendation ITU-T P.1502. This Recommendation also defines extensions of testing methodology and introduces a new type of tool to assist in testing. The methodology is designed to be easily extended to other use cases in future revisions of this Recommendation. It is important to understand that this Recommendation only covers the methodology for tests done from an individual user's (end-to-end) perspective, acting within a given DFS ecosystem under current load conditions in PSP, NPCI, Banks and network service providers. It may be desirable to extend the scope of testing to capacity tests, which would involve creation of defined load scenarios to a DFS ecosystem to determine the robustness of DFS functionality under these conditions. Such extensions can be easily created from the methodology described in this Recommendation.

2. References

The following ITU-T Recommendations and other references contain provisions which, through reference in this text, constitute provisions of this Recommendation. At the time of publication, the editions indicated were valid. All Recommendations and other references are subject to revision; users of this Recommendation are therefore encouraged to investigate the possibility of applying the most recent edition of the Recommendations and other references listed below.

A list of the currently valid ITU-T Recommendations is regularly published. The reference to a document within this Recommendation does not give it, as a stand-alone document, the status of a Recommendation.

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|----------------|---|
| [ITU-T E.804] | Recommendation ITU-T E.804 (02/2014), <i>QoS aspects for popular services in mobile networks</i> |
| [ITU-T P.1502] | Recommendation ITU-T P.1502 (01/2020), <i>Methodology for QoE testing of digital financial services</i> |

[ITU-T G.1033] Recommendation ITU-T G.1033 (10/2019), *Quality of service and quality of experience aspects of digital financial services*

3. Definitions

3.1. NPCI has provided the definitions for the following to use in UPIs mobile application payment system ;

NPCI National Payments Corporation of India (NPCI), an umbrella organisation for operating retail payments and settlement systems in India, is an initiative of Reserve Bank of India (RBI) and Indian Banks' Association (IBA) under the provisions of the Payment and Settlement Systems Act, 2007, for creating a robust Payment & Settlement Infrastructure in India.

PSP Payment service provider works with the bank to acquire new customers and facilitate payment.

SDK (for UPI application) A Mobile UPIs SDK is **a software package that contains a set of tools that can help to build platform-UPI mobile applications and implement new features in existing mobile apps**. The resources contained in an SDK include libraries, documents, code samples, guides, and APIs.

VPA **Virtual Payment Address unique account number specific to PSP and Bank linked with customer mobile provided by PSP. Virtual Payment Address provided by PSP. For example, <name>@psp. <name> could be provided by either PSP or customers can choose as per their preference.**

3.2. Definitions from ITU documents

Quality of experience (QoE) [b-ITU-T P.10]: The degree of delight or annoyance of the user of an application or service.

Quality of service (QoS) [b-ITU-T E.800]: The totality of characteristics of a telecommunications service that bear on its ability to satisfy stated and implied needs of the user of the service.

Digital Financial Service: Methods to electronically store and transfer funds; to make and receive payments; to borrow, save, insure, and invest; and to manage a person's or enterprise's finances; and where these services are accessed via mobile communication devices.

4. Abbreviations and acronyms

This Recommendation uses the following abbreviations and acronyms:

DFS Digital Financial Service

DAL Device Assignment List

IAP – In-Application Payment

KPI Key Performance Indicator

NPCI – National Payments Corporation of India

NIPL [NPCI International Payments Limited](#)

P2P Person-to-Person

P2M. Person-to-Merchant
PSP - Payment Service Providers
RBI Reserve Bank of India
SDK Software Development Kit
TAL Team Assignment List

5. Conventions

None

6. Test scenario under consideration

The basic scenario under consideration is the “Person-to-Person” (P2P) and “Person-to-Merchant” (P2M) money transfer or payment against service(s)/good(s) in the following cases;

- Money transfer or payment made between two parties in the same country, but using same DFS providers or PSP(in India) (inter-operator scenario)
- Money transfer or payment made between two parties in the same country, but using different DFS providers or PSP(in India) (inter-operator scenario)
- Money transfer between two parties in different countries, using the same DFS providers or PSP (e.g. national branches of a multi-national network operator)
- Money transfer between two parties in different countries, using different DFS providers.

It is important to state that the methodology for these variants is the same since in all these cases money or payment is transferred/made between two entities. There may be differences in the details of an operating sequence; these are, however, not greater than differences between same-operator, same-country operating sequences.

The P2P or P2M basic scenario and its modelling is described in detail in Recommendation ITU-T G.1033 and Recommendation ITU-T P.1502. For the sake of convenience of reading, the essential parts are explained here while for more detail, the reader is kindly referred to above mentioned Recommendations.

NOTE: In India, the DFS services might be registered under the central bank i.e., Reserve Bank of India (RBI) and money transfers considered to be bank transfers. From an end-to-end perspective, this is still a P2P or P2M transfer; technically, received money will in this case appear to come from that particular entity rather than from the actual sending entity.

6.1. Registration, roles, entities and action/event flow

6.1.1. Registration of mobile application by the user as follows.

Step -I:

- Download the PSP (*Payment Service Provider*) app (GooglePay, PhonePay, etc.)
- Register
- Add Account details
- Generate UPI PIN

** For every PSP app, the mobile SDK is provided by NPCI only.*

STEP – II

For Hard Binding or Device Finger Printing Process (acts as a First-Factor authentication in UPI)

- Download the PSP app and send an SMS for mobile number verification and ensure that you're performing this action with the same mobile number that is registered with the bank.

STEP – III

Customer Registration

- Download the PSP app and select your preferred bank
- Now the request will be sent to the PSP server and it will forward the same request to the NPCI (of the same mobile number)
- Now NPCI UPI server will forward the request to the Issuer Bank
- Bank will retrieve the account details to verify whether the number is linked to that person or not
- Now, the UPI will pass the same info to the PSP server

**PSP stores the IFSC and Account Number of the customer to the mobile apps (including device information)*

- Now, the customer will get all the linked bank accounts of his/her triggered SMS number and from there he/she can choose the preference
- PSP will now create the VPA to proceed ahead.

STEP – IV

PIN Generation

- From your Mobile application, select the option to Generate a PIN
- Now, the PSP server will request an OTP of that bank account to NPCI
- NPCI will forward the same request to the Issuer Bank
- The OTP will be forwarded back to the customer
- The customer will now be required to enter the last 6-digit debit card number (along with the expiry date and OTP)

** The performed action will be securely captured by NPCI SDK*

- PSP will now forward the OTP validation request
- UPI will decrypt the details (including PIN) from the PSP key and encrypt with the Issuer key
- Now, the Issuer bank will decrypt the data and will start validating the details (such as debit card no.) and OTP and will store the PIN

** Neither the PSP server nor the NPCI will have the authentication to save the PIN.*

6.1.2. Role of PSP

PSP works with the bank to acquire new customers, to onboard through mobile application after authentication through banks and NPCI and facilitate payment.

Their primary work is to offer front-end mobile applications to the customer and works closely with NPCI and banks. They also ensure the whole ecosystem of the transaction flow goes flawless

PSPs types

- Banks
- Third-Party Applications

A bank account can have multiple VPA handles-

- **Example:** HDFC Bank has multiple VPA's. **HDFC App** – @hdfc, **GooglePay** – @okhdfc, **PhonePay** – @ybl

PSP can have multiple Bank handles-

- **Example:** GooglePay. **AXIS Bank** – @okaxis, **HDFC** – @okhdfc, **ICICI** – @okicici, **SBI** – oksbi

QR – Abbreviated as “quick response”. It's a metric that holds information horizontally and can perform both horizontal and vertical action.

6.1.3. Entities

In each and every real-time transaction, certain parties are involved as the **4 Party Model through PSP application**, the models or entities and its responsibilities are:

1. **Payer PSP**
 - Customer onboarding
 - To create a UPI ID
 - Create device binding (first-factor authentication)
2. **Payee PSP**
 - On-board customer/merchant
 - facilitate money transfer/payment to the recipient using UPI
3. **Remitter Bank**
 - Hold & Debit Bank account for the transaction
 - Store and verify UPI PIN
4. **Beneficiary Bank**
 - Process incoming credits and funds into the beneficiary account
5. **NPCI**
 - Facilitate money transfer/payment to the recipient using UPI between payer and payee PSP
6. **Bank Account holders**
7. **Merchants**

Note: NPCI is the gateway interface for payee PSP and payer PSP

6.1.4. Transaction Flow

There are generally 2 varieties of transactions PUSH and PULL, below we will understand the PUSH method that is majorly used in the market.

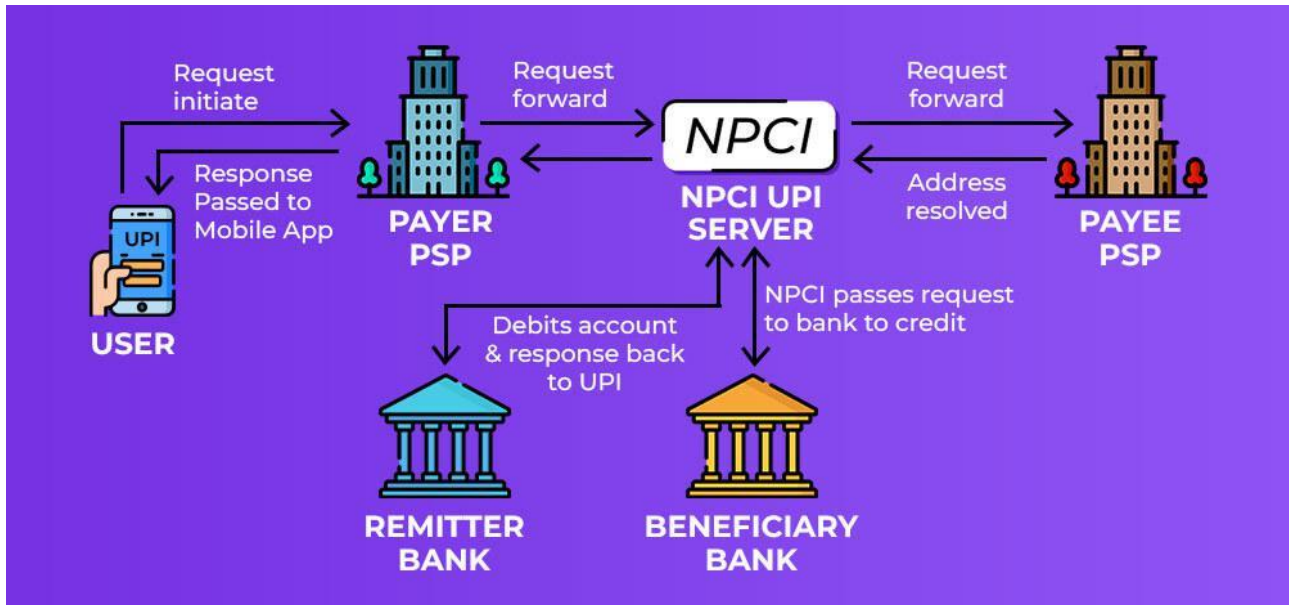


Figure 1 basic flow diagram of UPI payment system.

Phase – I

- The customer initiates the transaction either with Payee's mobile number, or VPA / QR
- Now, the Payer PSP will forward the same request to the NPCI
- Following that, the NPCI UPI server will forward the same request to the Payee's PSP for address resolution and authorization
- The payee PSP resolves the address and provides the account details (works with the Remitter bank)

Phase – II

- The Payee PSP will provide the bank details to UPI and the same will be forwarded to NPCI
- Now, the NPCI will check with the remitter bank to debit funds from the payer's account
- Once money gets debited, a credit request is sent to the beneficiary's bank
- The beneficiary bank credits the Payee's account and later responds to NPCI UPI
- Now, the NPCI UPI server passes the response to the status of the transaction via Payer's PSP to the customer

6.2 Test parameterization and neutral starting state

A particular property of systematic service tests is a frequency of service usages which is significantly higher than the usage frequency created by a typical end user.

While a high testing frequency leads to a high yield of samples for computation of QoS KPI, it is conceivable that the system has a certain “dead time” after each transaction, where the system would not accept a new transaction or create unexpected results of a transaction attempted within this period of time. It is advisable to be aware of this possibility and obtain respective information before actual parameters of a test campaign are determined.

The testing frequency can be controlled by a pause between transactions, which also acts as a guard time to allow the service under test to reach its neutral state again. Respective considerations are in full analogy to testing of e.g. telephony. A testing campaign, therefore, should contain a pre-testing phase with systematic tests to make sure that usage frequencies typical for testing do not affect testing results with respect to the end-user perspective.

NOTE: In actual implementations, there may be limitations applied by the DFS providers limiting the number of transactions per day, or the total amount of transferred money per day or another period of time. It is also conceivable that mechanisms exist which limit the frequency of testing. When setting up tests, it is important to check for such conditions. This starts with creating awareness of testers for effects of such mechanisms and be prepared to adapt data processing accordingly if respective effects are detected during a testing campaign. Otherwise, misclassifications may result, such as attributing effects of such limitation as functional failures of a Digital Financial Service. As the starting point for systematic testing, the guard time should typically be in the range of 10 to 30 seconds.

When testing is done manually, it is assumed that the system can handle all testing speeds which can be realized by human testers, as even an experienced tester will not work significantly faster than an experienced regular user of PSP/DFS. Therefore, no special requirements to slow down testing are applied.

In fully automated testing, it would also be possible to use the high degree of repeatability of such control to determine the appropriate guard time by probing, i.e. by systematically varying the guard time and check for respective effects.

There is a second category of effects which need to be considered, namely the possibility of a service-specific local memory (analogously to a browser's cache) which stores information related to previous transactions. The effect would be that in subsequent transactions, such information would be read from local memory instead of obtaining them by an over the air request to the service. This could then impact related measurement values or KPI.

As long as effects are quantitative rather than qualitative, it may not practicable and is not necessarily required to exclude frequency-dependent effects entirely. However, respective effects need to be recorded and documented carefully as part of the reporting in order to understand their impact on the testing conditions.

6.3 Re-initialization after unsuccessful transactions

If a transaction fails, in particular after a time-out condition has occurred, it shall be ensured that the service and the device or application are in the typical neutral starting state again, i.e. that no memory of previous error states remains in the system. So that customer end status of transactions would be clear.

6.4 Unsuccessful transactions without reversal of debited money

6.6 Automation of tests, periodically monthwise QoS and QoE of PSP for self network and rest of the network (operator wise with various combinations)

7 Transaction Model and DFS KPI

8 Creating the use case model from actual use case examples

9 Data Sources

9.1 Basic Considerations for data collection and processing

9.2 Data Structure Overview

9.3 Naming and formatting conventions

9.4 Naming and formatting conventions for data objects

9.5 Local log sheets

9.6 Team Assignment List (TAL)

9.7 Device Assignment List (DAL)

10 Special procedures in the field

10.1 Operational protocol if A and B party do not communicate directly

11 Testing Modes

11.1 Basics

11.2 Synchronized and asynchronous testing mode

11.3 Dual-function set-up and operation

12 Recording events of DFS test cases

12.1 Basic Functionality

12.2 Practical Application

13 Measurements in the background

14 General considerations about errors in measurements (API based data collection)

15 Data validation and processing

15.1 Overview

15.2 Plausibility and Validity checks

15.3 Data Processing

15.4 Time Profile

16 Background testing of mobile networks/Banks links with PSPs and mobile applications

16.1 General considerations

16.2 Testing Tools

16.3 KPI

Bibliography

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